

Aug-2026 high-vax-rate fits: summary of the three models

Median [95% interval] across posterior draws (one deterministic simulation per draw).

- **No initial immunity:** the earlier high-vax fit ([model_config_2026-08-high-vax-rates.json](#)).
- **20% R, doses of S+SV:** 20% of every age group starts in R; daily doses = scheduled proportion × (S + SV).
- **20% R, doses of total population:** 20% starts in R; daily doses = scheduled proportion × total population, all given to S.

Hospitalizations averted: *infection protection only* = no vaccination → infection-protection-only scenario; *hospitalization reduction* = infection-protection-only → full baseline; *total* = no vaccination → full baseline (the sum of the two). Percentages are of the no-vaccination hospitalizations. Attack rate = $(N - R(0) - S(T) - SV(T)) / N$, as a share of the whole age group (excludes the initially recovered). Effective IHR = fitted IHR_scale × I_to_H_prop (unvaccinated) or × IV_to_H_prop (vaccinated).

Fitted parameters

Parameter	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
beta_baseline	0.03656 [0.03078 – 0.04275] (best 0.03786)	0.0388 [0.03301 – 0.04548] (best 0.03834)	0.03403 [0.03008 – 0.03949] (best 0.03231)
humidity_impact	0.595 [0.2543 – 0.9554] (best 0.8756)	0.6304 [0.2027 – 0.9725] (best 0.4192)	0.7006 [0.2969 – 0.9812] (best 0.5228)
seed_scale_E	1.531 [0.4614 – 4.48] (best 1.367)	2.107 [0.7094 – 5.743] (best 2.512)	2.928 [1.025 – 6.641] (best 5.385)
posterior draws / simulations per scenario	638 / 638	638 / 638	627 / 627

Doses delivered (baseline)

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
0	31,409 [31,190 – 31,518]	25,066 [24,871 – 25,160]	31,787 [31,787 – 31,787]
1-4	251,202 [249,542 – 252,051]	200,464 [198,913 – 201,218]	221,572 [220,524 – 222,234]
5-12	421,409 [416,976 – 423,781]	335,772 [331,584 – 337,870]	429,737 [429,737 – 429,737]

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
13-17	222,816 [220,695 – 223,972]	177,595 [175,555 – 178,620]	226,884 [226,884 – 226,884]
18-49	1,154,627 [1,139,790 – 1,162,143]	919,221 [906,052 – 925,830]	1,181,355 [1,181,355 – 1,181,355]
50-64	827,816 [819,445 – 832,044]	659,698 [652,202 – 663,451]	842,877 [842,877 – 842,877]
65+	868,876 [864,287 – 871,177]	693,708 [689,608 – 695,762]	876,988 [876,988 – 876,988]
All	3,778,182 [3,741,571 – 3,796,678]	3,011,544 [2,978,607 – 3,027,793]	3,811,200 [3,810,152 – 3,811,862]

Hospitalizations averted, infection protection only

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
0	82 [53 – 107]	79 [53 – 99]	101 [61 – 132]
1-4	472 [351 – 569]	421 [306 – 502]	581 [445 – 689]
5-12	426 [286 – 573]	380 [250 – 509]	525 [346 – 679]
13-17	206 [138 – 302]	183 [118 – 262]	250 [164 – 357]
18-49	1,332 [949 – 1,765]	1,197 [851 – 1,564]	1,542 [1,146 – 2,006]
50-64	1,464 [1,038 – 1,942]	1,320 [931 – 1,760]	1,762 [1,295 – 2,342]
65+	8,569 [6,624 – 10,407]	7,683 [5,823 – 9,258]	10,601 [8,156 – 12,935]
All	12,608 [9,872 – 14,997]	11,299 [8,561 – 13,403]	15,370 [12,126 – 18,459]

% of no-vax hospitalizations averted, infection protection only

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
0	69.0% [63.8% – 72.5%]	66.8% [61.2% – 70.3%]	73.1% [68.5% – 76.4%]
1-4	74.5% [70.0% – 77.6%]	72.7% [67.6% – 75.7%]	79.1% [75.2% – 81.8%]
5-12	71.8% [66.2% – 75.6%]	69.5% [63.4% – 73.2%]	75.8% [70.9% – 79.1%]
13-17	69.1% [63.0% – 73.2%]	66.6% [60.0% – 70.6%]	72.7% [67.3% – 76.4%]
18-49	64.0% [57.6% – 68.3%]	61.3% [54.4% – 65.6%]	67.1% [61.0% – 71.2%]

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
50-64	65.4% [59.5% – 69.4%]	62.9% [56.4% – 66.9%]	68.9% [63.3% – 72.7%]
65+	63.0% [57.1% – 67.0%]	60.5% [54.1% – 64.5%]	66.5% [61.0% – 70.3%]
All	64.1% [58.4% – 68.1%]	61.7% [55.3% – 65.5%]	67.7% [62.1% – 71.4%]

Hospitalizations averted, hospitalization reduction

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
0	1 [1 – 1]	1 [1 – 1]	1 [1 – 1]
1-4	8 [6 – 11]	8 [6 – 10]	11 [8 – 14]
5-12	6 [4 – 8]	6 [4 – 8]	8 [6 – 10]
13-17	3 [2 – 4]	3 [2 – 3]	4 [2 – 5]
18-49	17 [14 – 20]	17 [14 – 21]	22 [17 – 26]
50-64	27 [22 – 33]	28 [22 – 33]	36 [29 – 44]
65+	909 [825 – 990]	905 [824 – 984]	1,208 [1,089 – 1,329]
All	971 [886 – 1,052]	968 [884 – 1,046]	1,289 [1,172 – 1,410]

% of no-vax hospitalizations averted, hospitalization reduction

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
0	0.6% [0.6% – 0.8%]	0.7% [0.6% – 0.8%]	0.7% [0.7% – 0.9%]
1-4	1.3% [1.1% – 1.5%]	1.4% [1.2% – 1.6%]	1.5% [1.3% – 1.7%]
5-12	1.0% [0.9% – 1.2%]	1.1% [1.0% – 1.3%]	1.2% [1.0% – 1.4%]
13-17	0.9% [0.7% – 1.0%]	0.9% [0.8% – 1.1%]	1.0% [0.9% – 1.2%]
18-49	0.8% [0.7% – 0.9%]	0.9% [0.8% – 1.0%]	0.9% [0.8% – 1.1%]
50-64	1.2% [1.1% – 1.4%]	1.3% [1.2% – 1.5%]	1.4% [1.2% – 1.7%]
65+	6.7% [6.0% – 7.7%]	7.1% [6.4% – 8.3%]	7.6% [6.7% – 8.9%]
All	4.9% [4.4% – 5.8%]	5.3% [4.7% – 6.2%]	5.7% [5.0% – 6.6%]

Hospitalizations averted, total

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
0	83 [53 – 108]	79 [53 – 100]	102 [61 – 133]

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
1-4	480 [357 – 580]	429 [313 – 511]	591 [454 – 703]
5-12	432 [292 – 581]	385 [256 – 516]	534 [351 – 689]
13-17	208 [140 – 305]	186 [120 – 265]	253 [167 – 362]
18-49	1,348 [965 – 1,783]	1,215 [864 – 1,582]	1,563 [1,165 – 2,031]
50-64	1,492 [1,063 – 1,975]	1,347 [952 – 1,791]	1,799 [1,328 – 2,381]
65+	9,481 [7,589 – 11,327]	8,581 [6,683 – 10,218]	11,835 [9,253 – 14,223]
All	13,584 [10,817 – 15,997]	12,284 [9,539 – 14,405]	16,663 [13,378 – 19,732]

% of no-vax hospitalizations averted, total

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
0	69.6% [64.6% – 73.0%]	67.5% [62.0% – 70.9%]	73.9% [69.4% – 77.0%]
1-4	75.8% [71.5% – 78.8%]	74.1% [69.3% – 77.0%]	80.6% [76.9% – 83.1%]
5-12	72.8% [67.4% – 76.5%]	70.6% [64.7% – 74.2%]	77.0% [72.4% – 80.2%]
13-17	70.0% [64.1% – 74.0%]	67.5% [61.1% – 71.5%]	73.7% [68.5% – 77.3%]
18-49	64.8% [58.6% – 69.0%]	62.2% [55.4% – 66.3%]	68.0% [62.1% – 72.0%]
50-64	66.6% [60.9% – 70.5%]	64.2% [58.0% – 68.0%]	70.3% [65.0% – 74.0%]
65+	69.7% [64.8% – 72.9%]	67.6% [62.4% – 70.9%]	74.1% [69.8% – 77.1%]
All	69.1% [64.1% – 72.5%]	66.9% [61.4% – 70.3%]	73.3% [68.7% – 76.4%]

Hospitalizations, baseline

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
0	36 [24 – 50]	37 [27 – 52]	36 [21 – 49]
1-4	155 [108 – 203]	151 [108 – 193]	143 [105 – 187]
5-12	162 [118 – 212]	163 [119 – 208]	158 [112 – 205]
13-17	91 [63 – 124]	90 [60 – 122]	91 [63 – 119]
18-49	740 [596 – 903]	738 [603 – 917]	739 [595 – 885]
50-64	753 [600 – 913]	756 [610 – 915]	760 [613 – 929]

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
65+	4,144 [3,748 – 4,506]	4,119 [3,766 – 4,476]	4,137 [3,720 – 4,534]
All	6,068 [5,628 – 6,525]	6,071 [5,648 – 6,481]	6,054 [5,640 – 6,507]

Hospitalizations, no vax

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
0	118 [77 – 156]	117 [79 – 151]	138 [83 – 179]
1-4	636 [476 – 773]	579 [432 – 697]	734 [563 – 885]
5-12	593 [422 – 785]	547 [392 – 711]	695 [469 – 889]
13-17	299 [209 – 422]	274 [186 – 382]	345 [240 – 479]
18-49	2,080 [1,610 – 2,642]	1,946 [1,509 – 2,428]	2,303 [1,811 – 2,880]
50-64	2,233 [1,680 – 2,871]	2,101 [1,584 – 2,690]	2,562 [1,995 – 3,273]
65+	13,591 [11,519 – 15,562]	12,684 [10,542 – 14,497]	15,974 [13,171 – 18,587]
All	19,615 [16,830 – 22,221]	18,348 [15,339 – 20,553]	22,761 [19,404 – 25,931]

Attack rate, baseline

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
0	5.1% [3.7% – 8.0%]	5.0% [3.7% – 7.6%]	5.2% [3.8% – 7.3%]
1-4	5.0% [3.6% – 7.8%]	4.9% [3.6% – 7.4%]	4.7% [3.5% – 6.6%]
5-12	7.7% [5.6% – 12.0%]	7.5% [5.6% – 11.3%]	7.2% [5.4% – 10.3%]
13-17	8.7% [6.3% – 13.4%]	8.5% [6.3% – 12.7%]	8.4% [6.3% – 11.8%]
18-49	8.7% [6.3% – 13.4%]	8.5% [6.3% – 12.7%]	9.0% [6.7% – 12.6%]
50-64	7.4% [5.4% – 11.5%]	7.2% [5.4% – 10.8%]	7.6% [5.6% – 10.7%]
65+	4.8% [3.5% – 7.6%]	4.8% [3.5% – 7.2%]	5.2% [3.8% – 7.3%]
All	7.5% [5.4% – 11.6%]	7.3% [5.4% – 11.0%]	7.6% [5.7% – 10.8%]

Attack rate, no vax

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
0	16.6% [13.4% – 22.3%]	15.2% [12.4% – 19.5%]	19.2% [15.9% – 23.5%]
1-4	19.7% [16.0% – 26.1%]	17.8% [14.8% – 22.7%]	22.3% [18.8% – 26.9%]
5-12	27.6% [22.9% – 35.2%]	24.7% [20.8% – 30.6%]	30.0% [25.8% – 35.2%]
13-17	28.3% [23.5% – 36.1%]	25.3% [21.4% – 31.4%]	30.7% [26.4% – 36.0%]
18-49	24.3% [19.9% – 32.0%]	22.0% [18.3% – 27.8%]	27.3% [23.1% – 32.7%]
50-64	21.4% [17.4% – 28.5%]	19.5% [16.1% – 24.8%]	24.4% [20.5% – 29.6%]
65+	13.1% [10.5% – 17.9%]	12.0% [9.8% – 15.7%]	15.4% [12.7% – 19.1%]
All	22.0% [18.0% – 29.0%]	19.9% [16.6% – 25.3%]	24.8% [21.0% – 29.8%]

IHR, unvaccinated (I_to_H)

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
0	1.023% [0.601% – 1.365%]	1.127% [0.676% – 1.378%]	1.021% [0.593% – 1.375%]
1-4	1.170% [0.765% – 1.380%]	1.177% [0.744% – 1.385%]	1.198% [0.842% – 1.381%]
5-12	0.355% [0.209% – 0.511%]	0.366% [0.221% – 0.525%]	0.384% [0.233% – 0.524%]
13-17	0.252% [0.153% – 0.414%]	0.261% [0.153% – 0.413%]	0.272% [0.171% – 0.406%]
18-49	0.288% [0.176% – 0.422%]	0.298% [0.192% – 0.425%]	0.285% [0.195% – 0.400%]
50-64	0.731% [0.438% – 1.077%]	0.756% [0.472% – 1.134%]	0.732% [0.493% – 1.066%]
65+	8.527% [5.466% – 11.865%]	8.658% [5.648% – 11.802%]	8.490% [5.960% – 11.482%]
All	n/a	n/a	n/a

IHR, vaccinated (IV_to_H)

Age group	No initial immunity	20% R, doses of S+SV	20% R, doses of total population
0	0.933% [0.548% – 1.245%]	1.028% [0.616% – 1.257%]	0.931% [0.541% – 1.254%]
1-4	1.067% [0.697% – 1.258%]	1.073% [0.678% – 1.264%]	1.093% [0.768% – 1.260%]
5-12	0.324% [0.191% – 0.466%]	0.334% [0.201% – 0.479%]	0.350% [0.213% – 0.478%]
13-17	0.230% [0.139% – 0.377%]	0.238% [0.140% – 0.377%]	0.248% [0.156% – 0.370%]
18-49	0.262% [0.160% – 0.384%]	0.272% [0.175% – 0.388%]	0.260% [0.178% – 0.364%]
50-64	0.666% [0.399% – 0.981%]	0.689% [0.430% – 1.033%]	0.667% [0.449% – 0.971%]
65+	5.883% [3.771% – 8.187%]	5.974% [3.897% – 8.143%]	5.858% [4.112% – 7.923%]
All	n/a	n/a	n/a